

ITEM: {{ QUESTION_ID: PT2-Q05 }}
 DEMAND: {{ DIFFICULTY_BADGE: MEDIUM }}
 TARGET GRADE: GRADE 9 MATH

CONCEPT: {{ PRIMARY_CONCEPT: Coordinate Transformations }}
 LINK: {{ CORE1_LINK: Core1 Lesson A1 }}
 SRC: {{ SOURCE_REF: PT2-2023-TERM2 }}

{{ QUESTION_TITLE: Line with Slope 2/3 Passing Through Point (4, -1) }}

ORIGINAL EXAM QUESTION / ATTEMPT

{{ QUESTION_STEM }}:
 A straight line L passes through point A(4, -1) with slope $m = 2/3$.
 Which of the following points also lies on line L, and what is the y-intercept of L?

 Options:
 (a) Point (-2, -5) and y-intercept = $-11/3$ (b) Point (1, -3) and y-intercept = $-11/3$
 (c) Point (7, 1) and y-intercept = $11/3$ (d) Point (2, -2) and y-intercept = $-8/3$

H1-H3 HINT LADDER (READ ONE AT A TIME ONLY IF STUCK)

* {{ H1_NOTICE }}: The slope $m = (y - y_1) / (x - x_1)$ is identical for ALL points on the line.
 * {{ H2_MODEL }}: Use point-slope form: $(y - (-1)) = 2/3 (x - 4) \Rightarrow 3(y + 1) = 2(x - 4)$.
 Rearrange into standard linear equation form: $2x - 3y - 11 = 0$.
 * {{ H3_START }}: Substitute candidate points into $2x - 3y - 11 = 0$ to see which gives 0.
 For y-intercept, evaluate at $x = 0$: $2(0) - 3y - 11 = 0 \Rightarrow y = -11/3$.

SEMANTIC REASONING ROUTE

Step 1 [INTERPRET]: Identify given geometric parameters: point (4, -1), slope $m = 2/3$.
 Step 2 [MODEL & BIND]: Convert to algebraic form using point-slope invariant: $2x - 3y - 11 = 0$.
 Step 3 [DISCRIMINATE]: Test options: $(-2, -5) \Rightarrow 2(-2) - 3(-5) - 11 = -4 + 15 - 11 = 0$. [VALID]
 Step 4 [VERIFY INTERCEPT]: Set $x = 0 \Rightarrow y = -11/3$. Select matching option (a).

VERIFIED SOLUTION & CHECK

{{ SOLUTION_FINAL_ANSWER }}: Correct Option is (a).

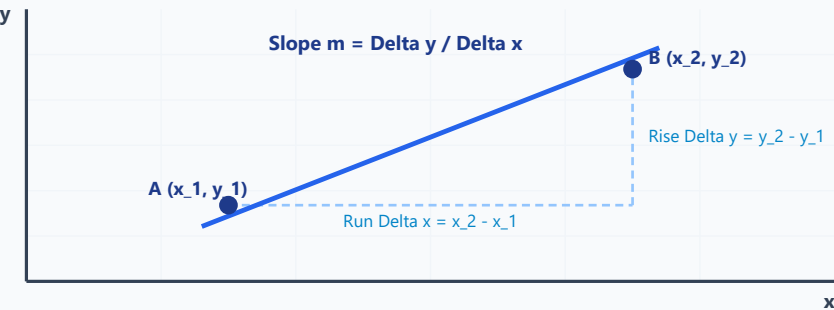
Execution Walkthrough:
 1. Standard Equation: $2x - 3y - 11 = 0$.
 2. Option (a) Point Check: Substitute (-2, -5):
 $LHS = 2(-2) - 3(-5) - 11 = -4 + 15 - 11 = 0 = RHS$. (Satisfies L)
 3. Intercept Determination:
 L crosses y-axis where $x = 0 \Rightarrow 2(0) - 3y - 11 = 0 \Rightarrow -3y = 11 \Rightarrow y = -11/3$.

{{ VERIFICATION_CHECK }}:

Compute slope between given point A(4, -1) and discovered point B(-2, -5):
 $m_{calc} = (-1 - (-5)) / (4 - (-2)) = 4 / 6 = 2/3$. Exactly matches problem statement.

Permanent Invariant: Points lie on a line if and only if they satisfy its equation.

{{ RECONSTRUCTED_SOURCE_FIGURE: 2D Line L on Cartesian Grid }}



{{ H0_ATTEMPT_WORKSPACE: Work Here * Try Before Looking at Hints }}

Your Final Choice: [] Reason: _____